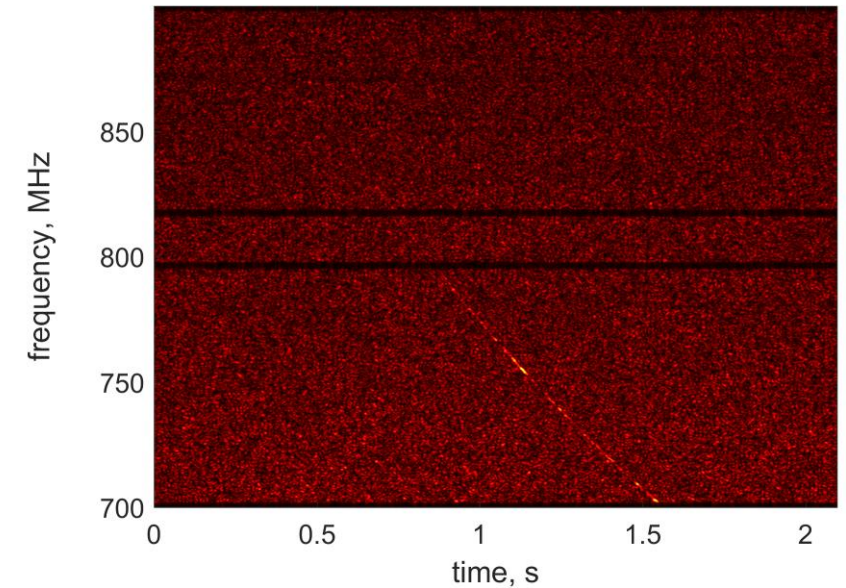
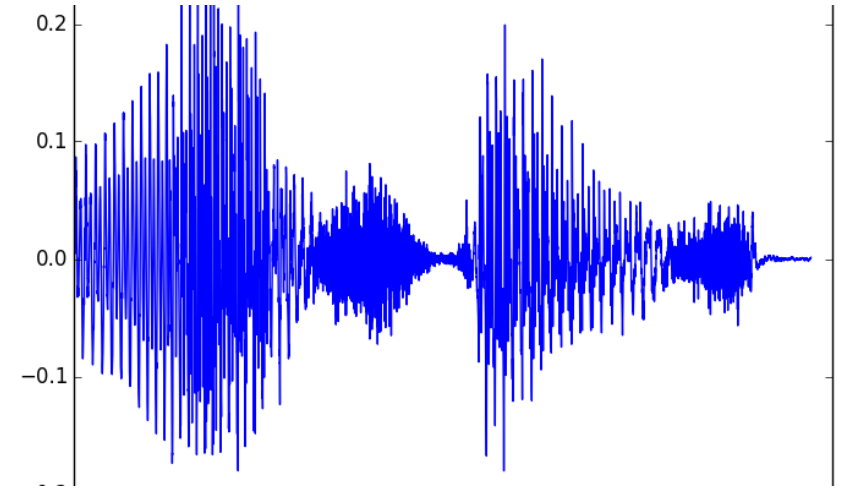


Signal Processing for Radio Astronomy Applications



Signals

- Signals = functions in 1D, 2D, 3D, etc.
- Examples: audio signal, images, video signal
- Dependence on time and frequency (or space and spatial frequencies)



Systems

- Cellular antenna
- Camera
- Tomographer (CT, MRI, etc.)
- Optical telescope
- Radio telescope
- Radar
- Any other receiver or antenna



Operations on Signals

As with any mathematical functions,
signals can be manipulated:

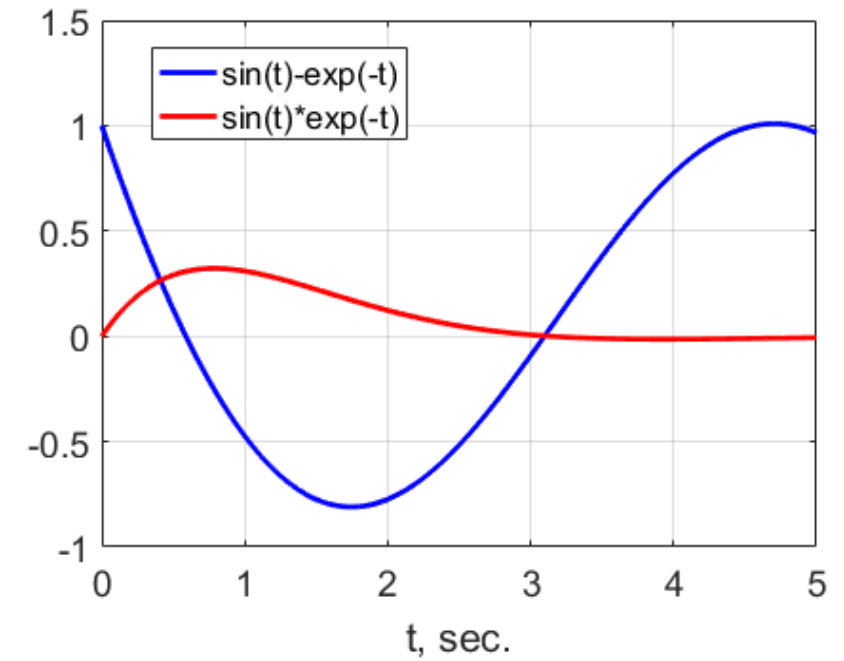
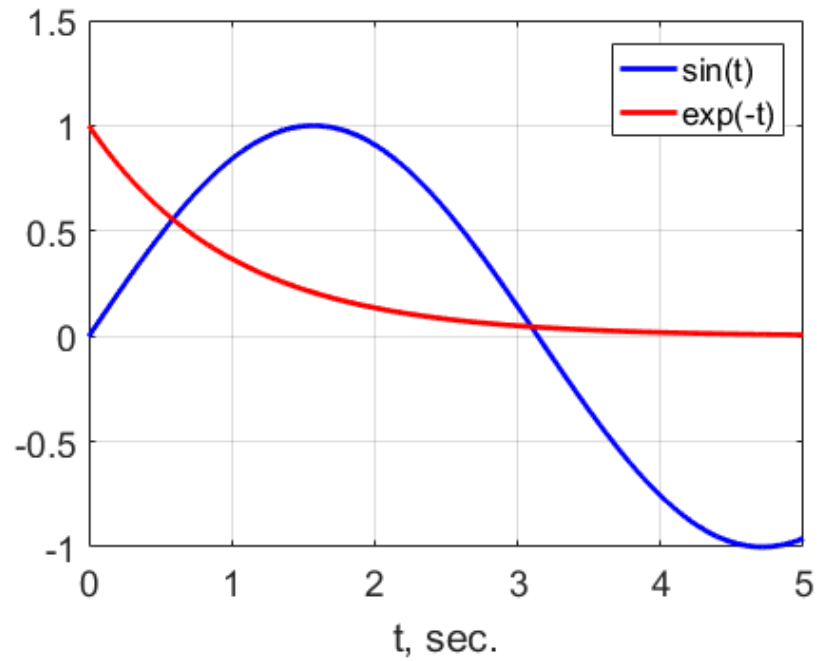
Adding and subtracting signals

Multiplying signals

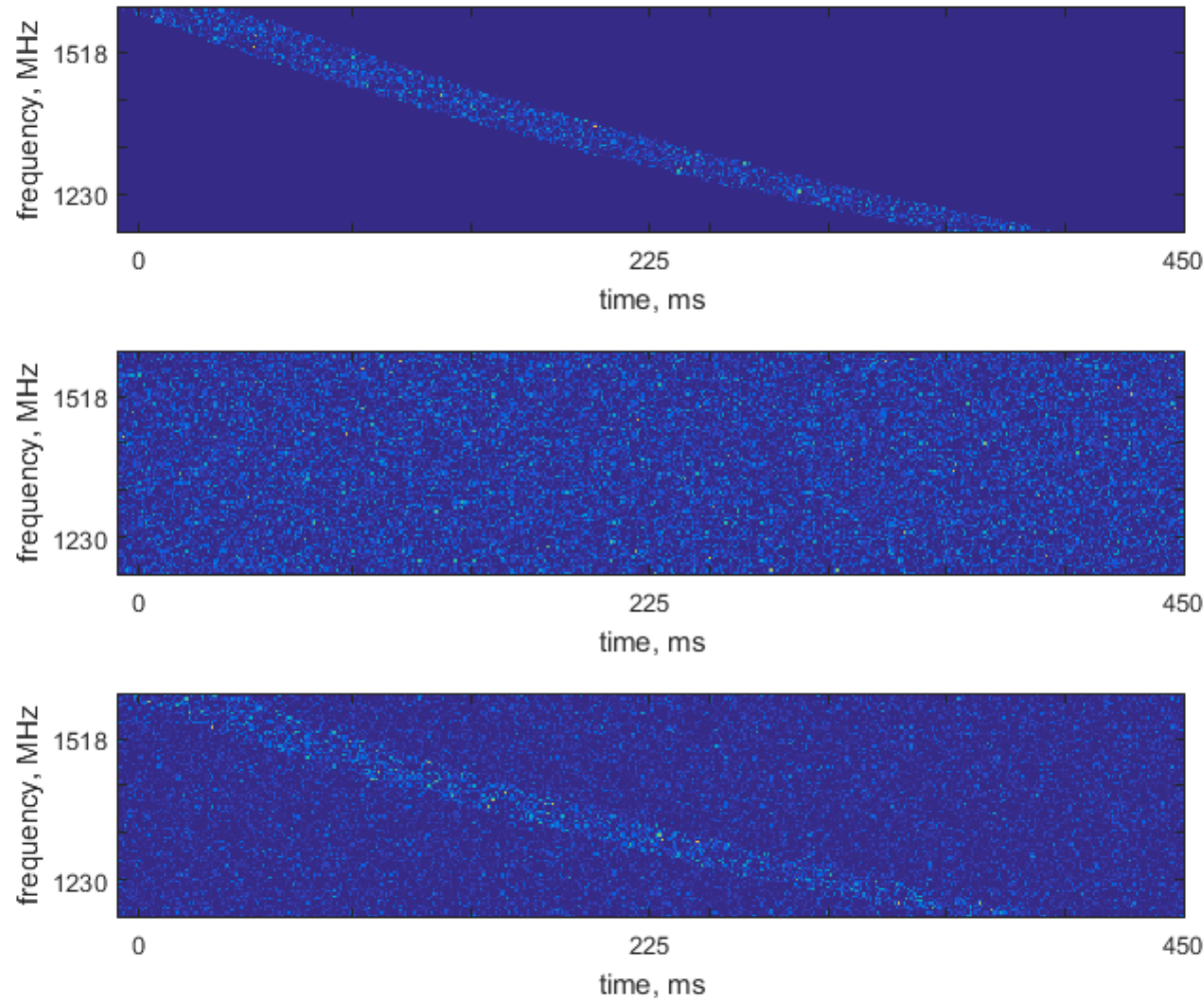
Scaling in time or frequency

Delaying

Correlating / convolving



Adding / Subtracting / Multiplying



Adding signals:
radio astronomy

Simulated Lorimer
burst

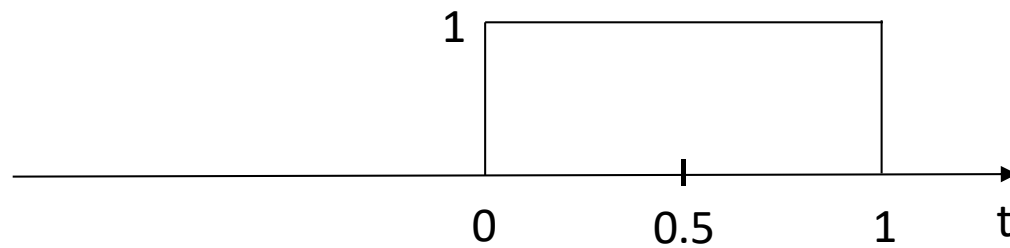
$DM = 375$

$S/N = 25$

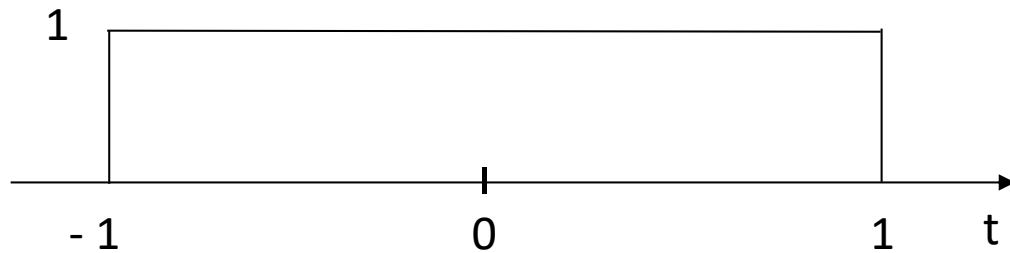
Time delay and scaling



$$\Pi(t) = \begin{cases} 1, & \text{if } |t| \leq 0.5 \\ 0, & \text{if } |t| > 0.5 \end{cases}$$



$$\Pi(t - 0.5)$$



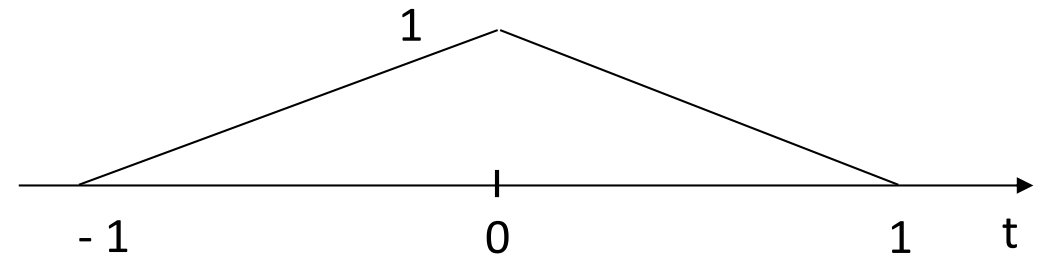
$$\Pi(0.5t)$$

Correlation of signals

- $g(t)$ and $h(t)$ are two real valued signals
- Correlation of $g(t)$ and $h(t)$ is defined as

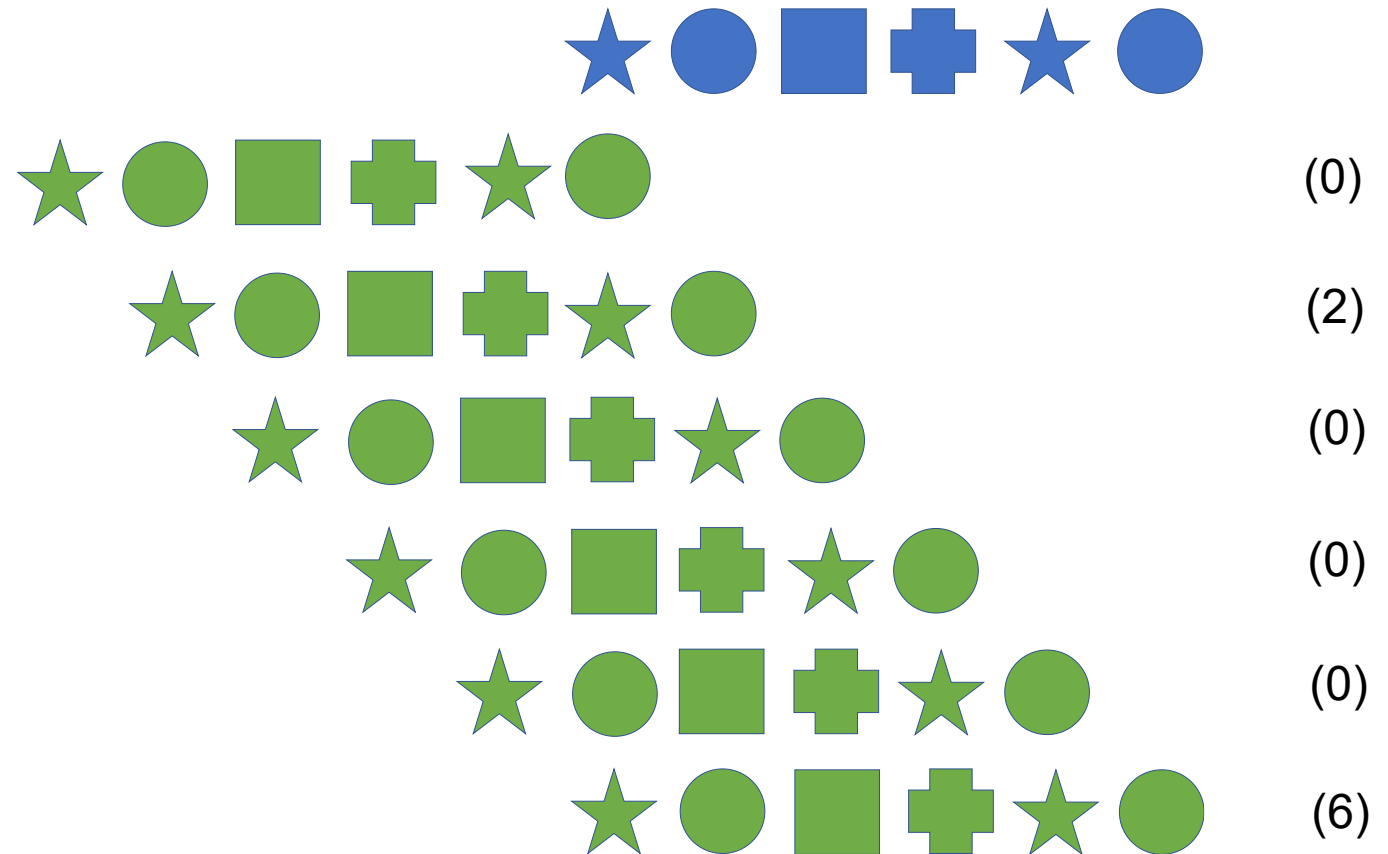
$$R(t) = g(t) * h(t) = \int_{-\infty}^{+\infty} g(\tau)h(t + \tau)d\tau$$

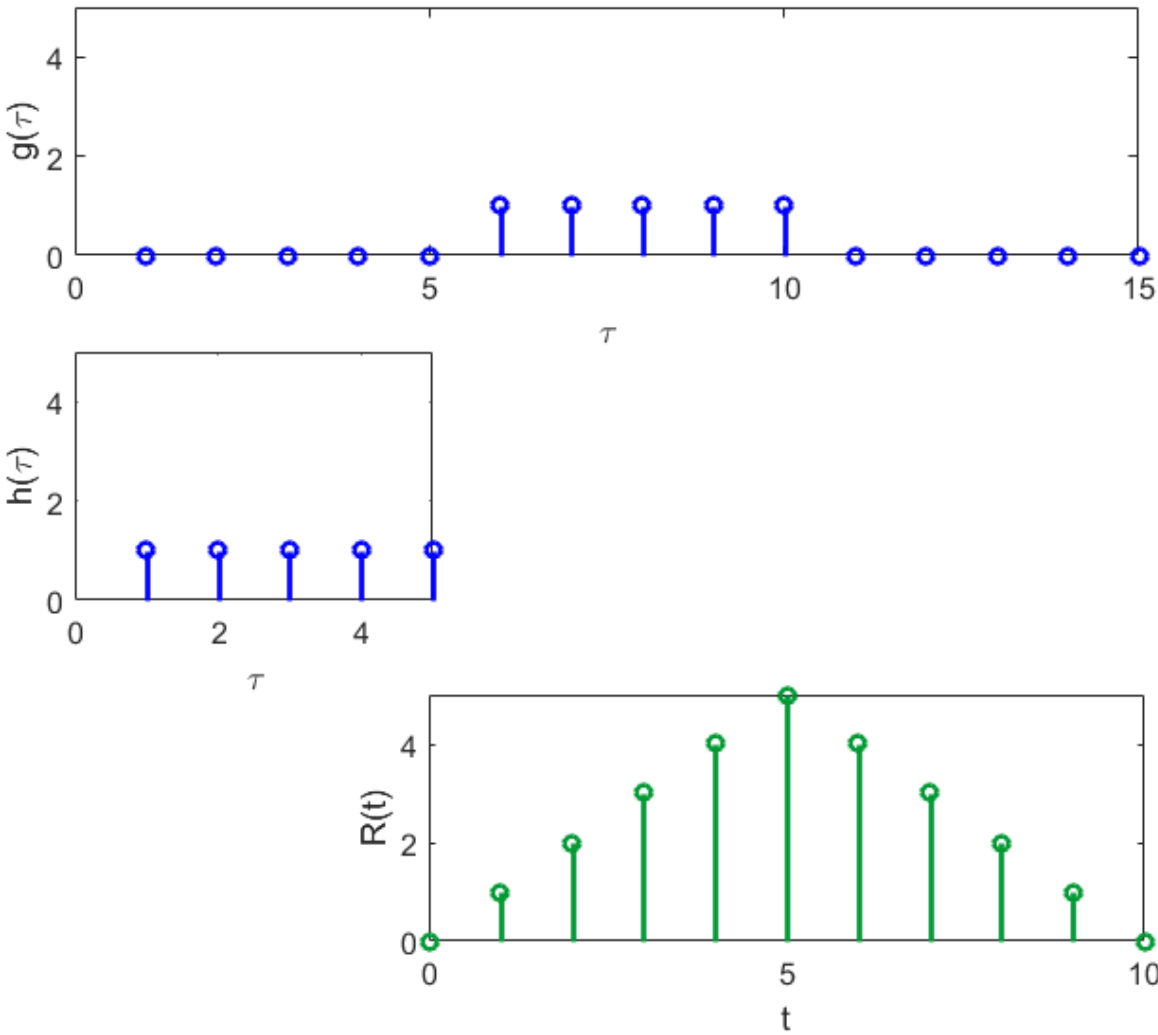
- Example: $\Pi(t) * \Pi(t)$



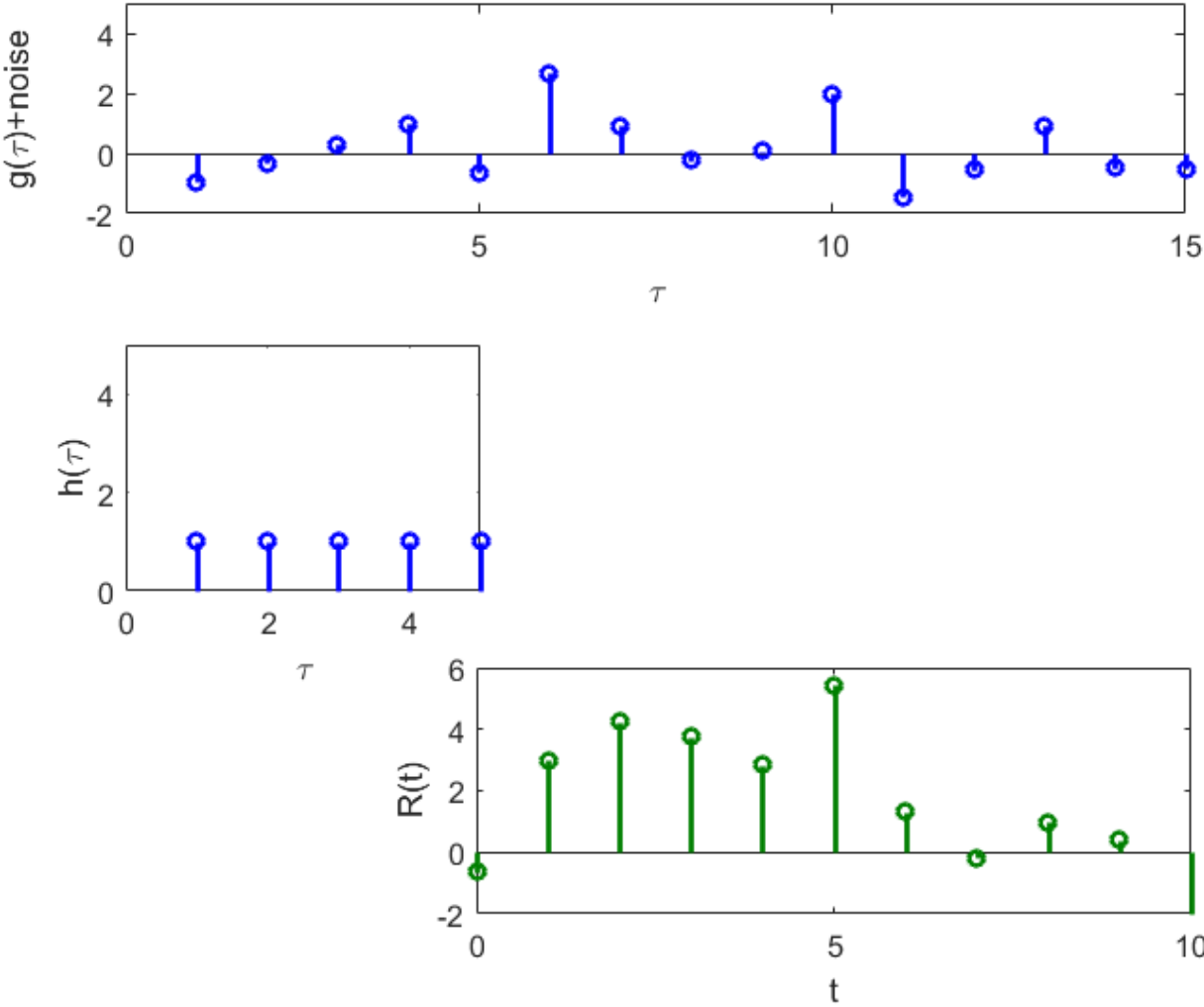
Why correlation?

- It's like matching patterns
- Operators: shift, logical comparison
- Assign one if two objects match (zero, otherwise), add all ones, find the maximum score. This is the best match.
- The same idea is used to search for signals.



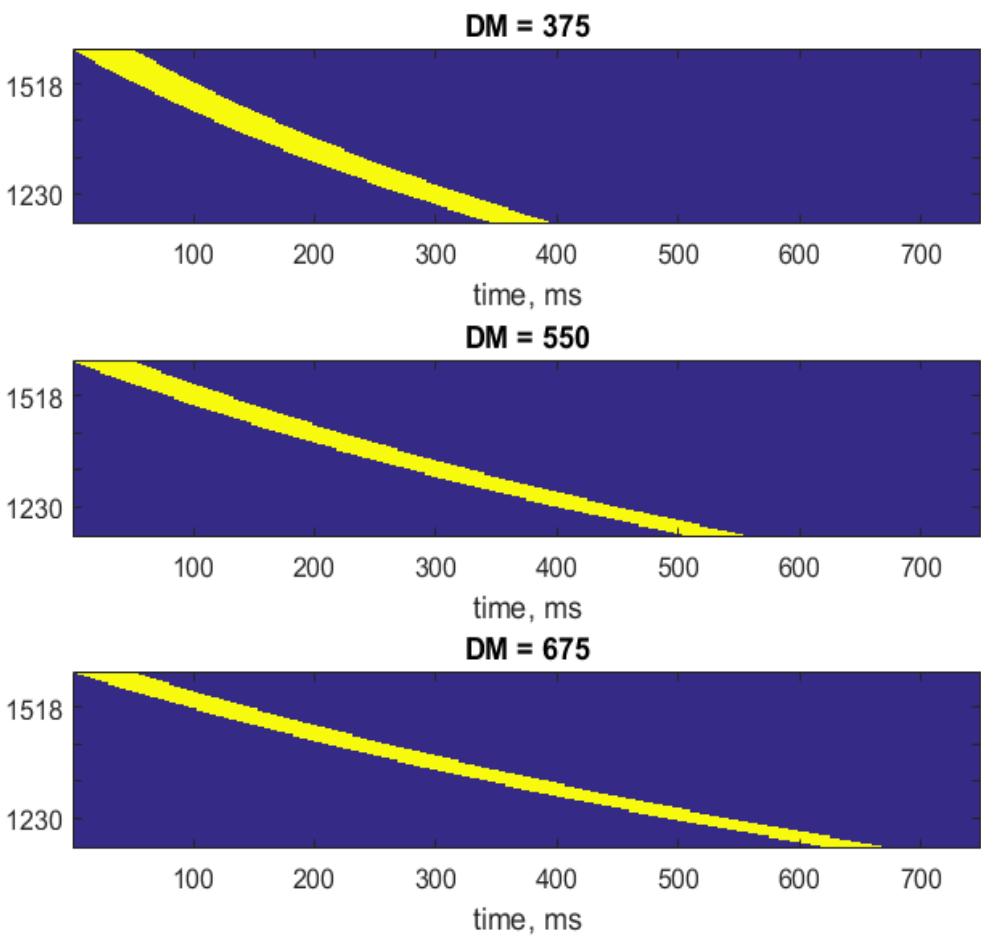
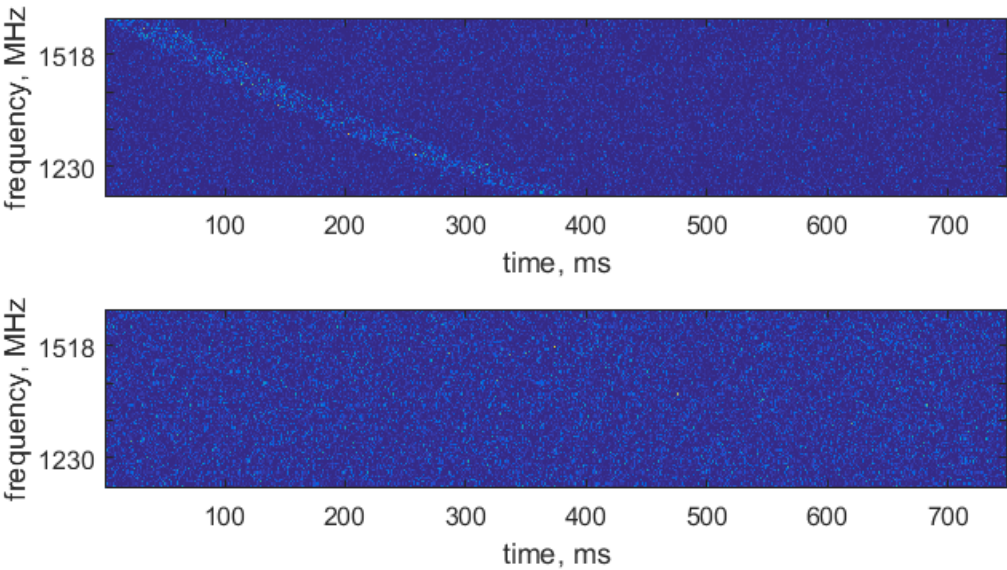


Illustration



Noisy signal

Matching signals in 2D



Signal + Noise	Noise only
1984.6	726.8
1013.0	740.3
896.8	755.8